Developer Toolkit: Terminal, Git & Python Basics

Topics to be covered:

- CLI — examples
- Git Commands
- Basic Python Programming.

1 Navigation Commands

Purpose	Linux / PowerShell	CMD	Example
Show current folder	`pwd`	`cd`	`pwd`
List files	`ls`	`dir`	`ls`
Change directory	`cd folder`	`cd folder`	`cd project`
Go back	`cd ..`	`cd ..`	`cd ..`
Clear terminal	`clear`	`cls`	`cls`

2 Folder Commands

Purpose	Linux / PowerShell	CMD	Example
Create folder	`mkdir test`	`mkdir test`	`mkdir demo`
Remove empty folder	`rmdir test`	`rmdir test`	`rmdir demo`

3 File Commands

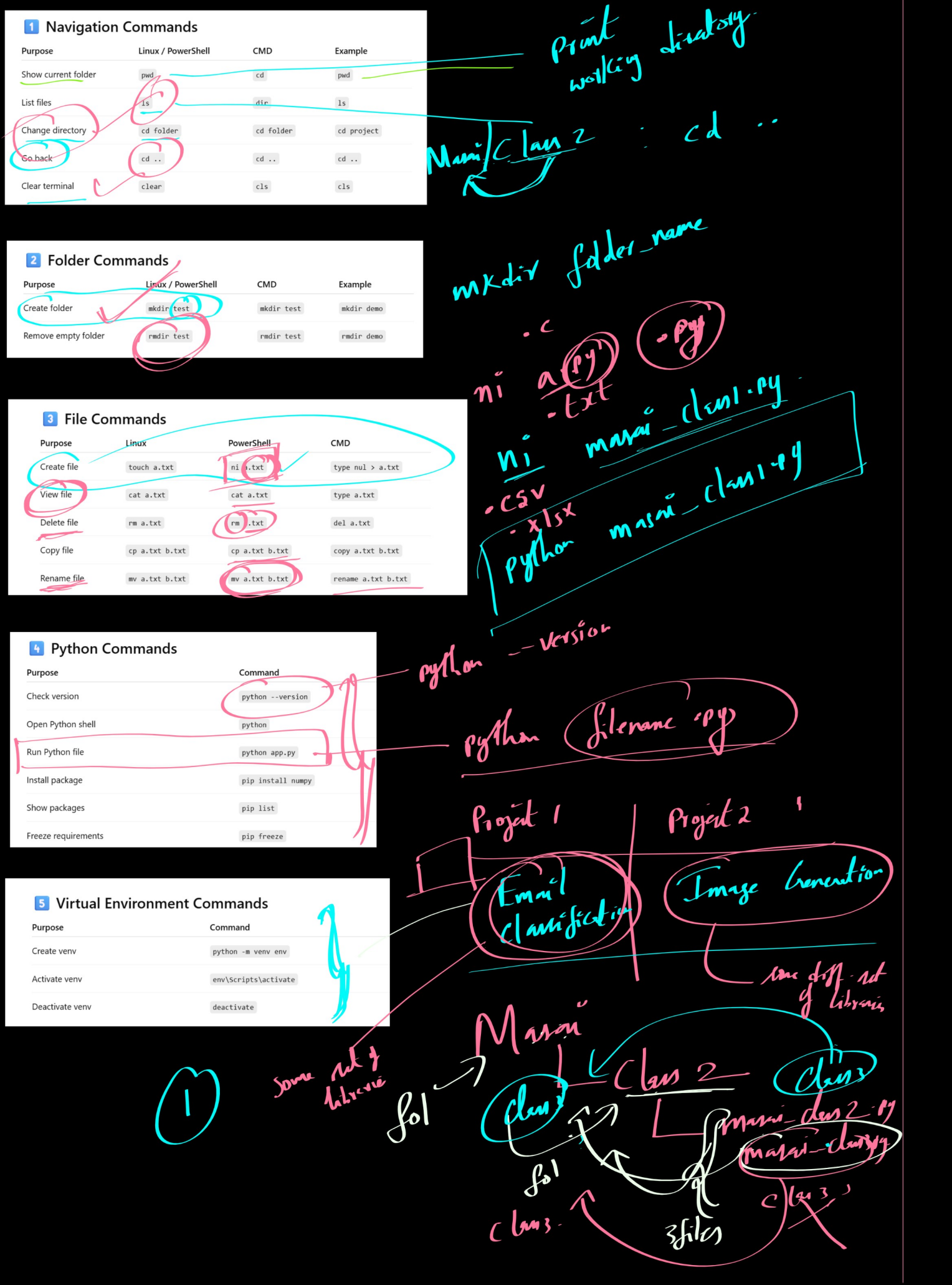
Purpose	Linux	PowerShell	CMD
Create file	`touch a.txt`	`ni a.txt`	`type nul > a.txt`
View file	`cat a.txt`	`cat a.txt`	`type a.txt`
Delete file	`rm a.txt`	`rm a.txt`	`del a.txt`
Copy file	`cp a.txt b.txt`	`cp a.txt b.txt`	`copy a.txt b.txt`
Rename file	`mv a.txt b.txt`	`mv a.txt b.txt`	`rename a.txt b.txt`

4 Python Commands

Purpose	Command
Check version	`python --version`
Open Python shell	`python`
Run Python file	`python app.py`
Install package	`pip install numpy`
Show packages	`pip list`
Freeze requirements	`pip freeze`

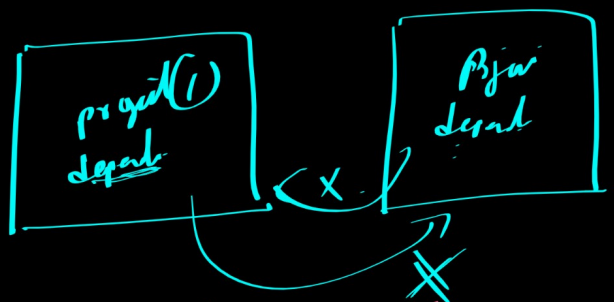
5 Virtual Environment Commands

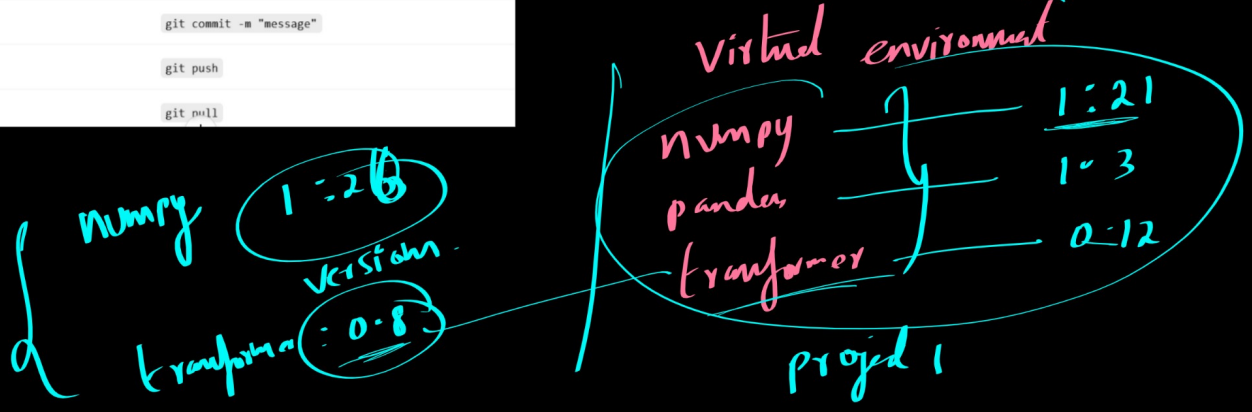
Purpose	Command
Create venv	`python -m venv env`
Activate venv	`env\Scripts\activate`
Deactivate venv	`deactivate`

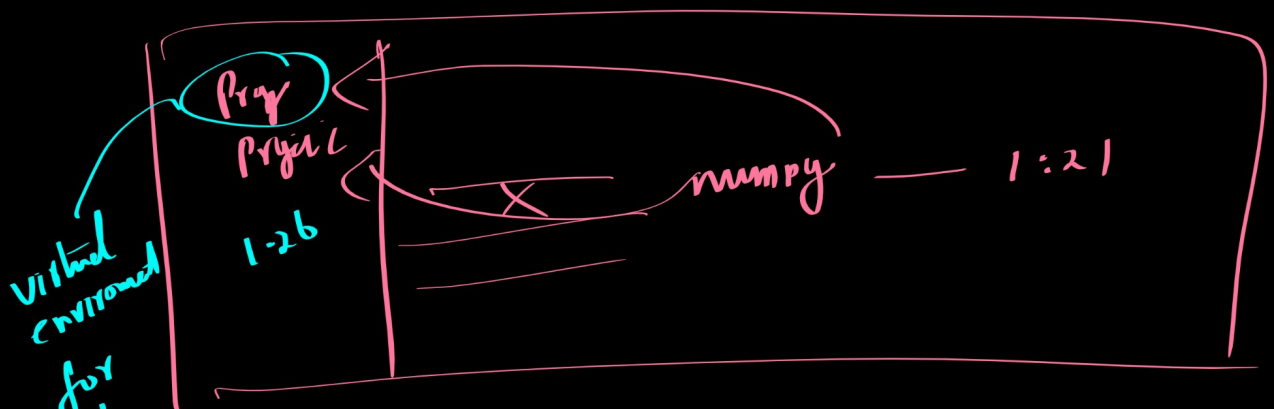


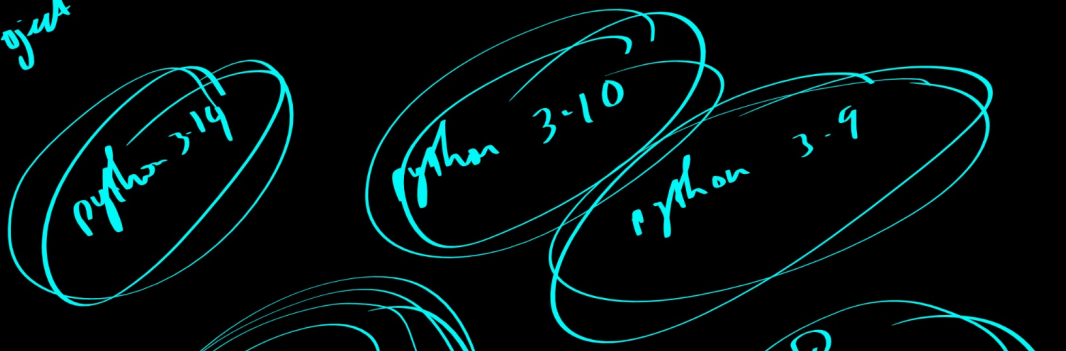
Git Commands (Very Important)

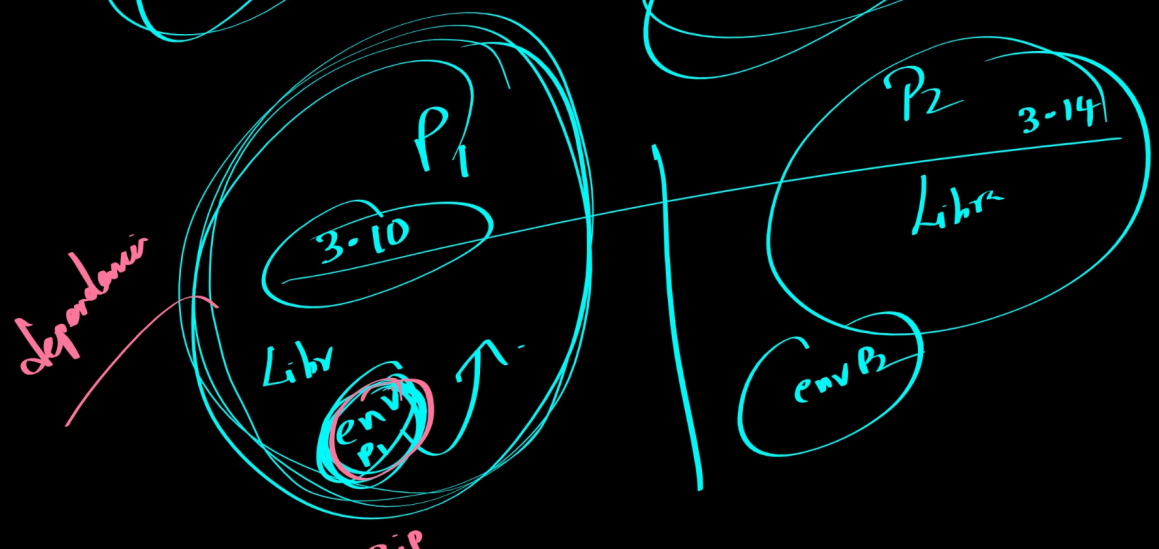
Purpose	Command
Initialize git	`git init`
Clone repo	`git clone <url>`
Check status	`git status`
Add files	`git add .`
Commit	`git commit -m "message"`
Push code	`git push`
Pull latest code	`git pull`

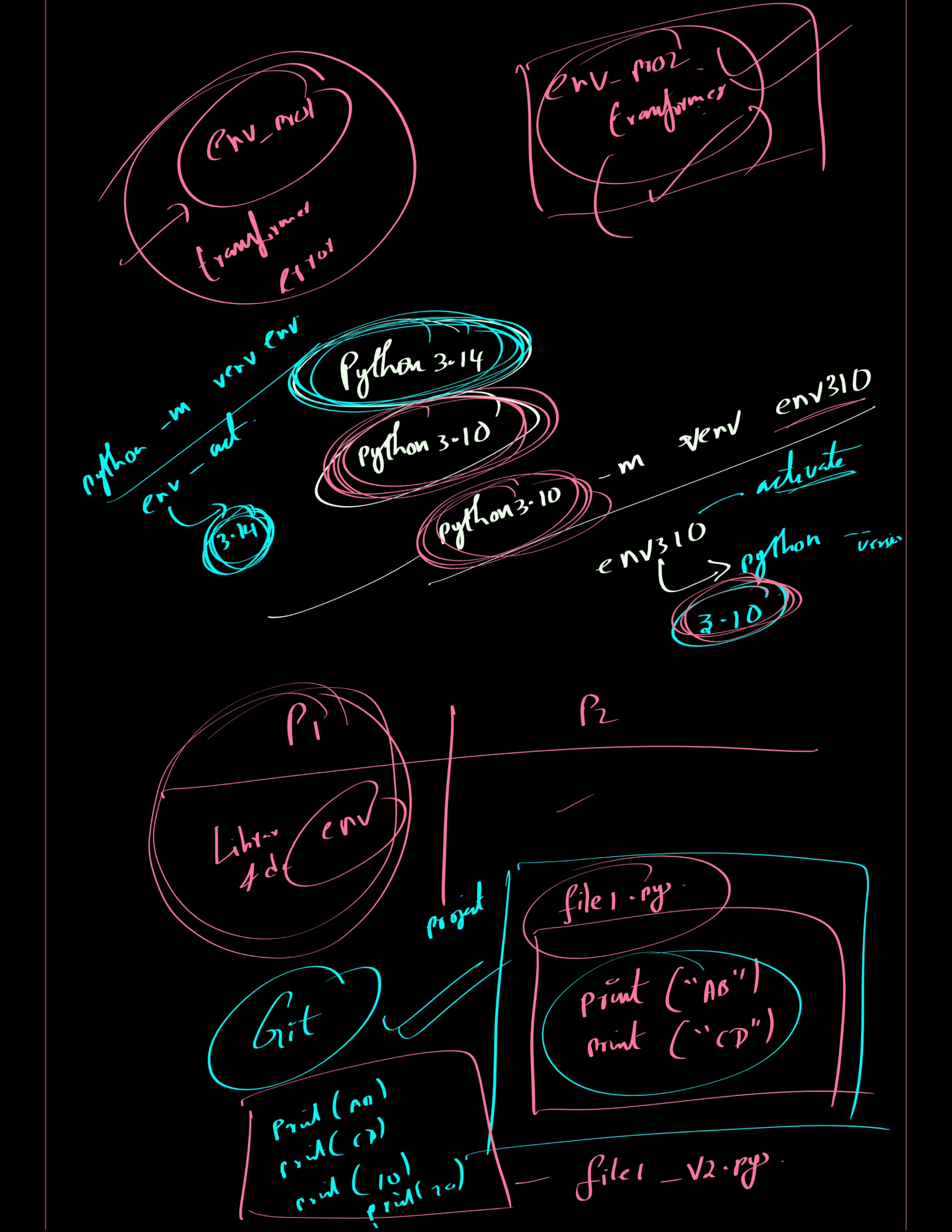




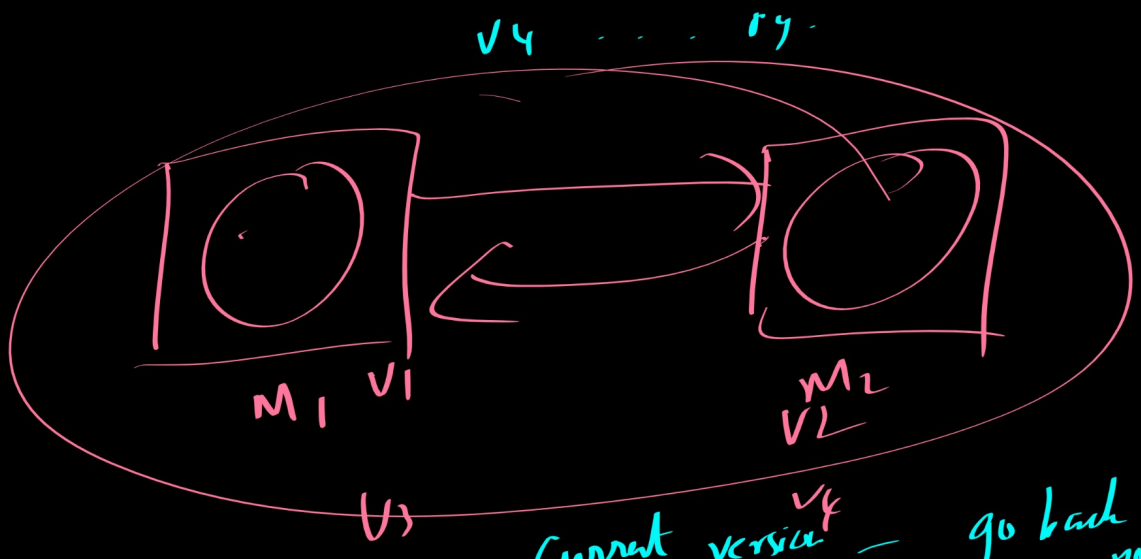




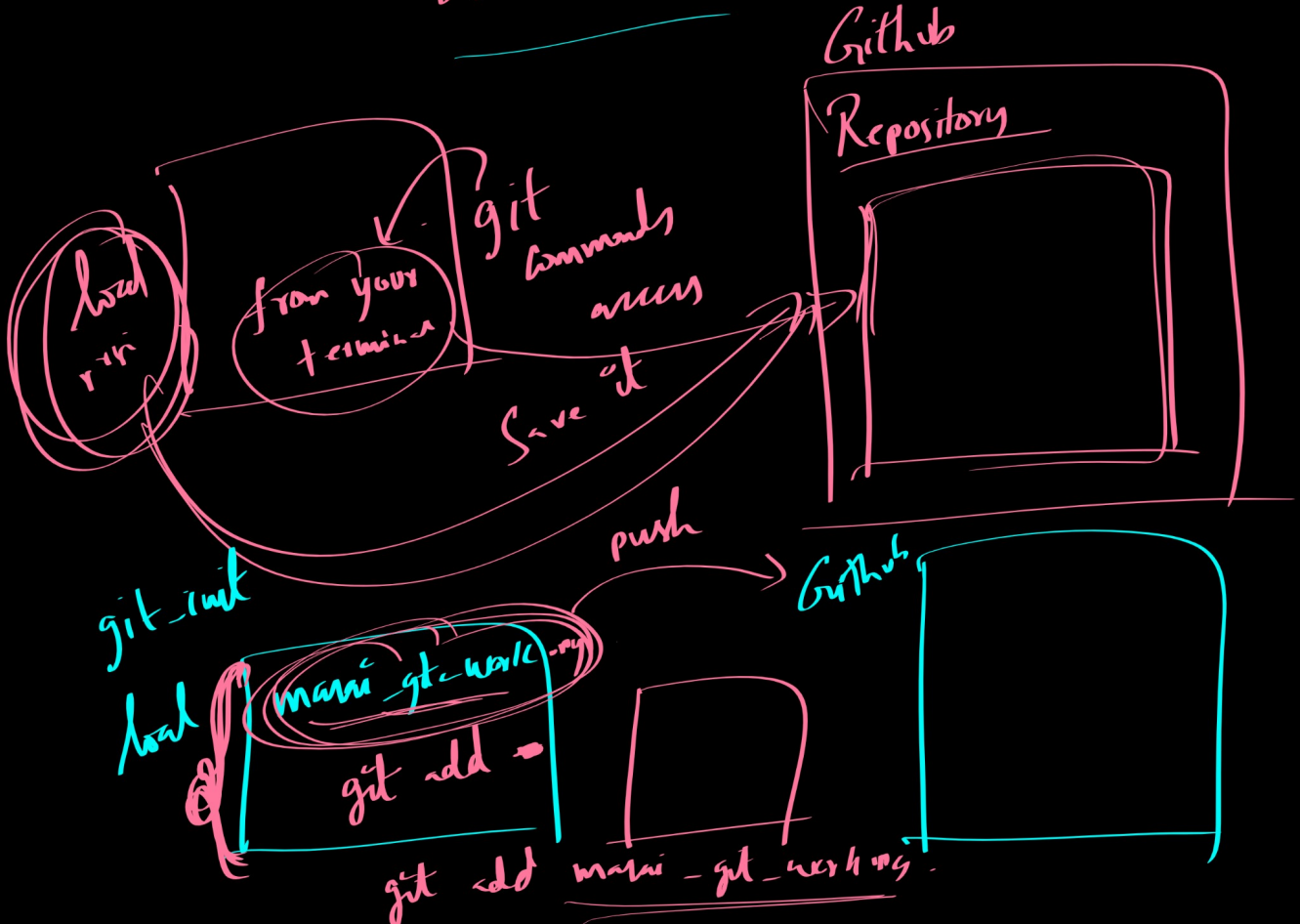


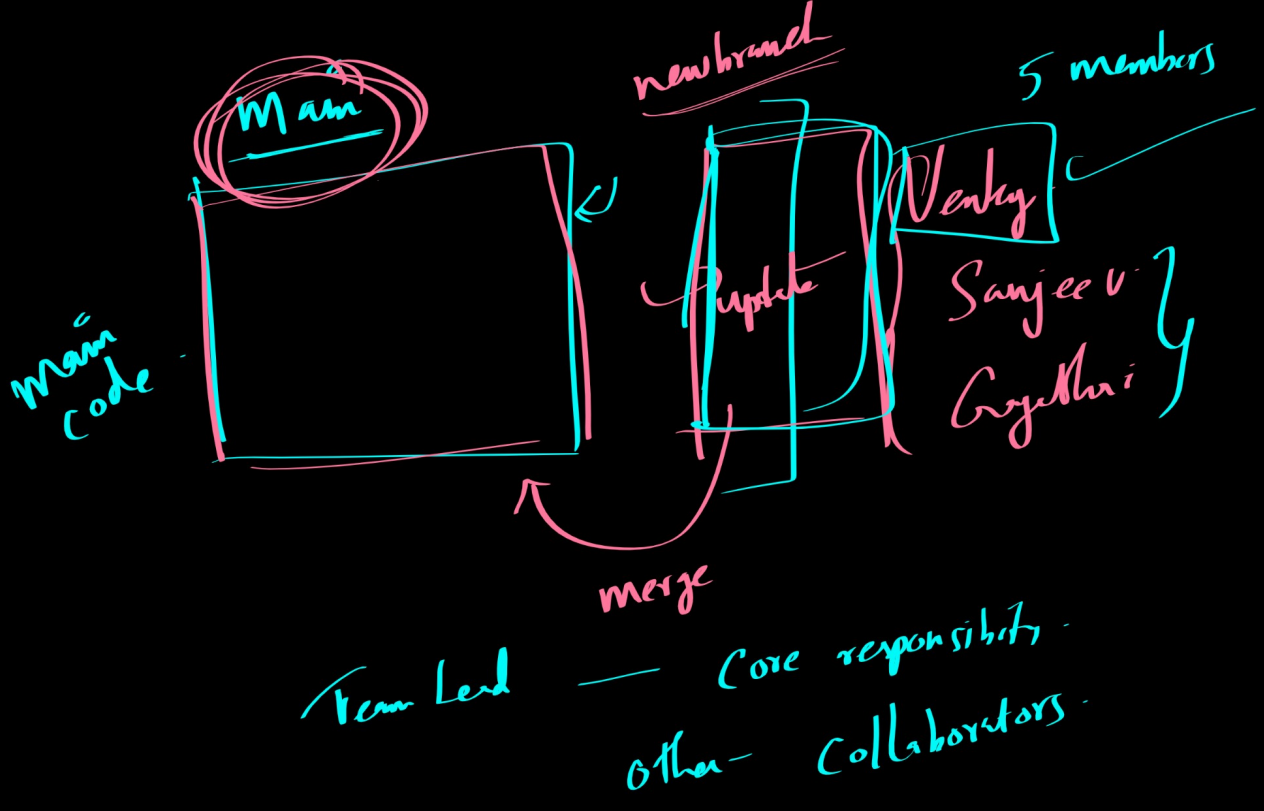

env_mol
transformer
env_mol2
transformers
python -m venv env
env - act
3.14
Python 3.14
Python 3.10
Python3.10
-m venv env310
activate
env310
python
3.10
P1
P2
Libr-
4de
env
Git
project
file1.py
print("AB")
print("CD")
print(aa)
print(23)
print(10)
print(20)
file1_V2.py

file1_v3.py

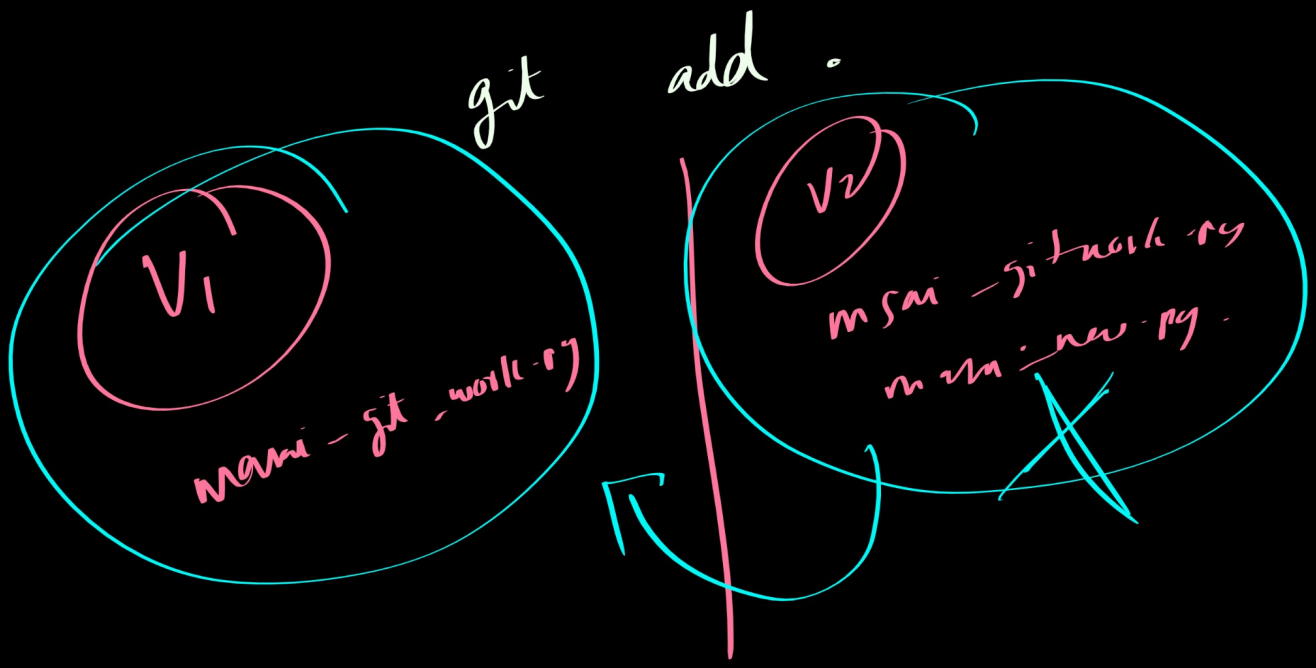


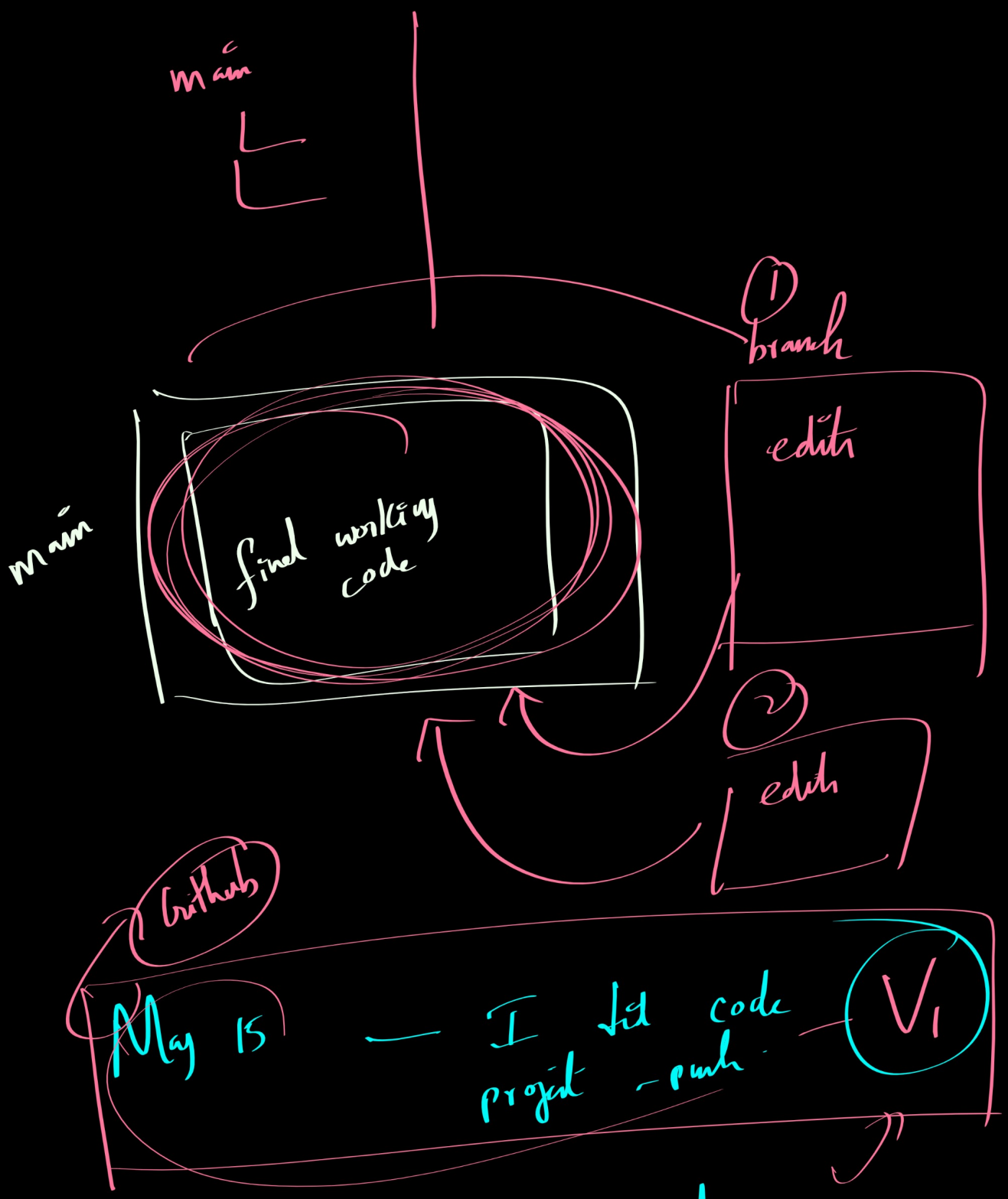
Git → Version control system that tracks changes in your project over time.





git clone https://github.com/venkateshelangovan/project1-masai.git




main
1
branch
edits
main
final working code
2
edits
Github
May 15 — I hit code
project - push
V1
May 16 — I - edit - V2
May 17 — I edit - V3

May 15

I want to go back to my interview

June

July 20 - I do all - V100

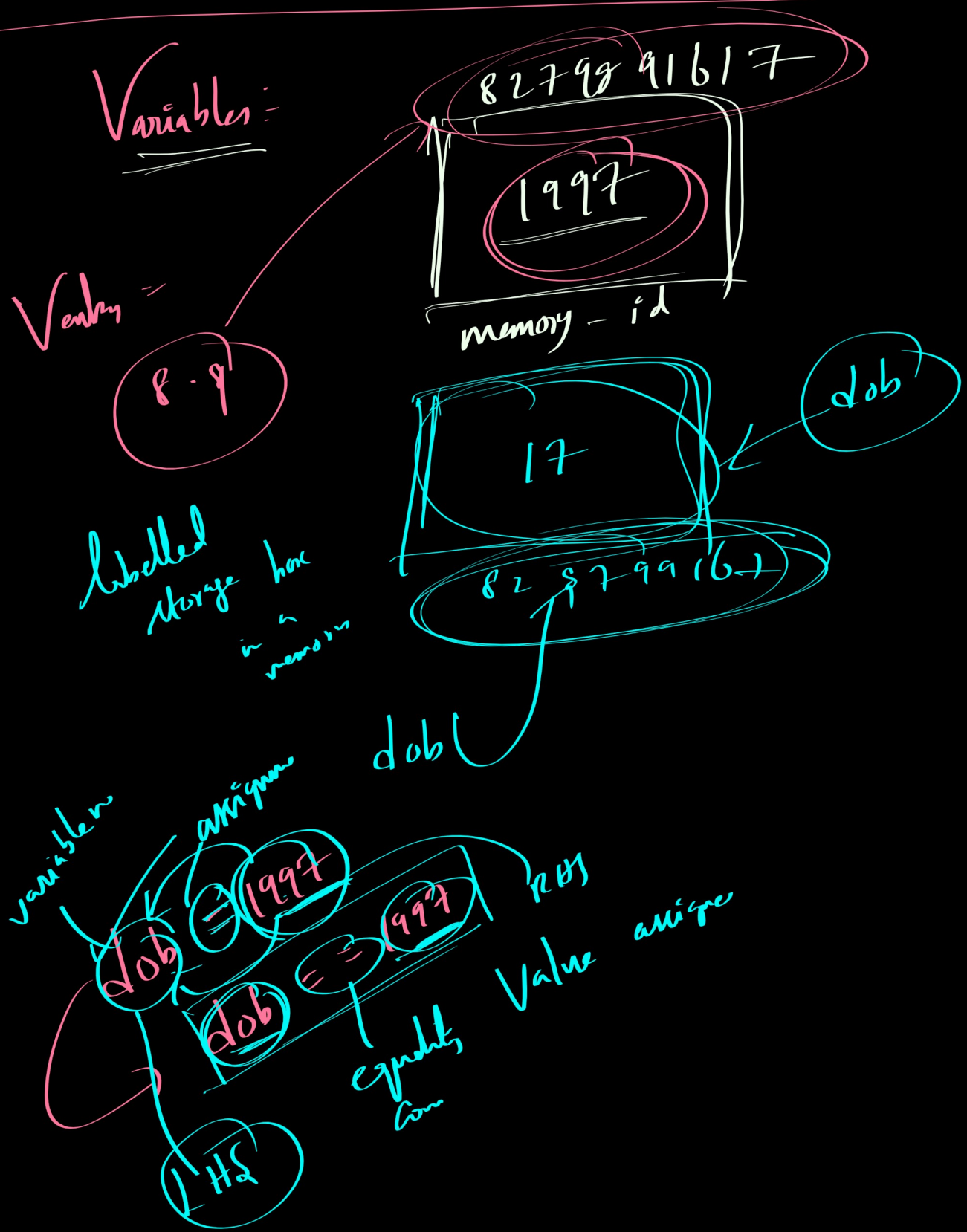
https://git-scm.com/install/windows

Python Programming

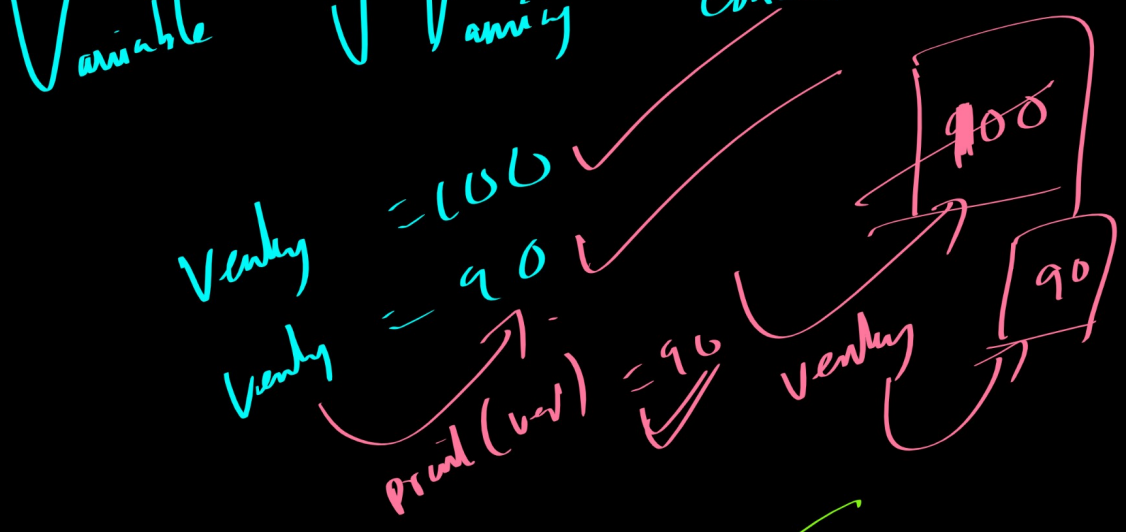
- work & calculations / decisions / build something

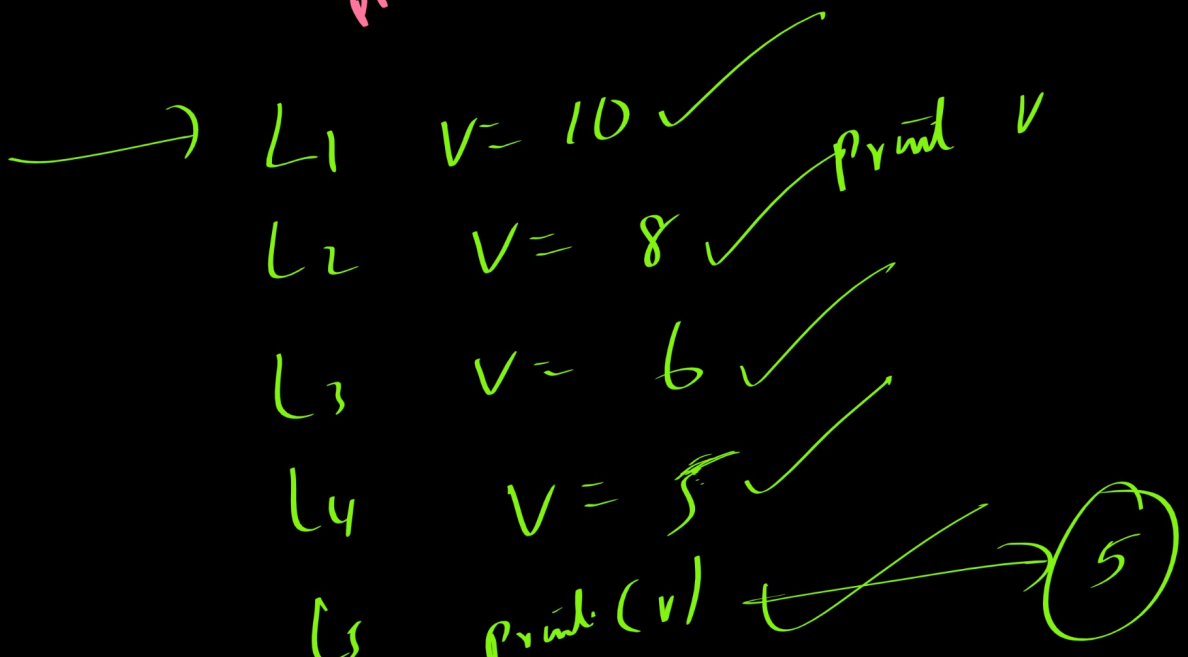
Store some data

Variables :

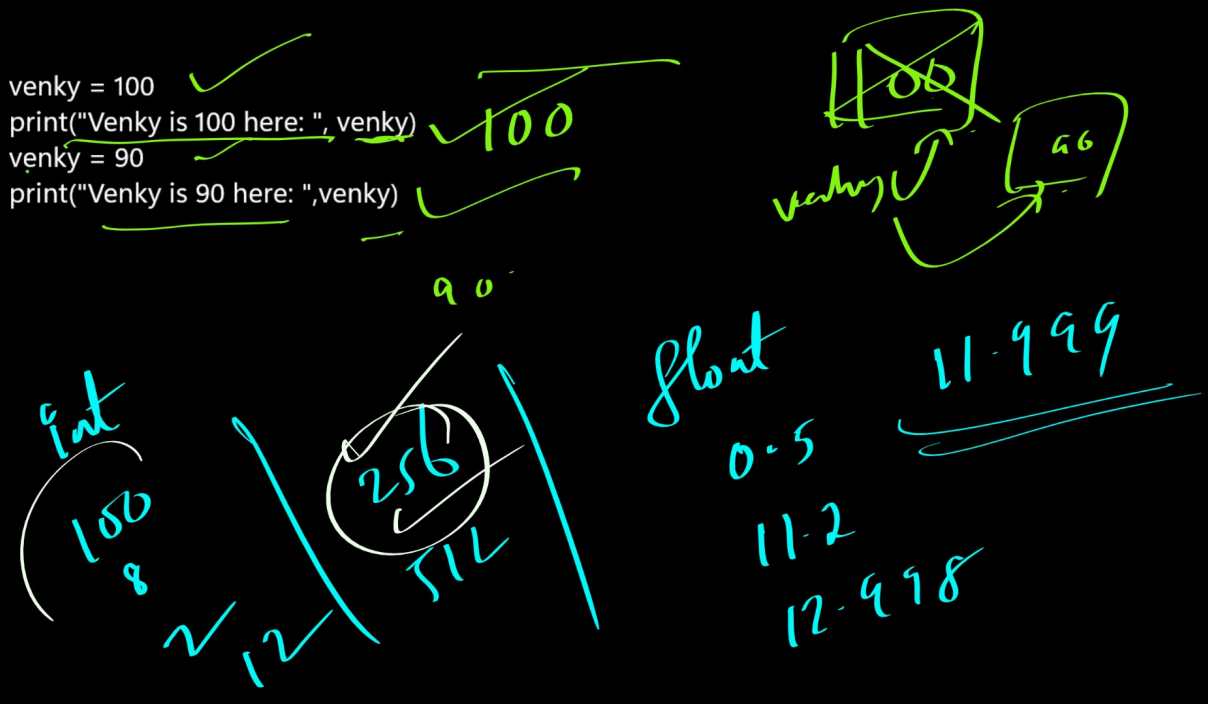


Variable Naming Conventions






```
venky = 100
print("Venky is 100 here: ", venky)
venky = 90
print("Venky is 90 here: ",venky)
```



256.0 — read it as float

"a", "b", "c", ... "z"

"Vashi", "IIIT Hyd"

Variable name

name_2 = "name_4"

value

name_2 = name_4

int, str, float.

```
if cgpa > 9:
    grade = S

else:
    grade = "Fail"
```

```
cgpa = 8.5

if cgpa > 9:
    print("Grade : S")
else:
    print("Grade : F")
```

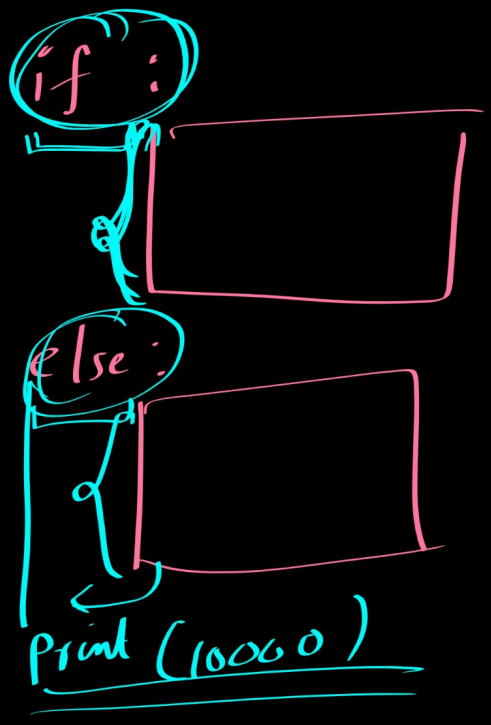
2Grade

2Grade

```
cgpa = 8.5
if cgpa > 9:
    print("Grade : S")
    print(100)
    print(1000)
else:
    print("Grade : F")
    print(-100)
    print(-1000)
print(10000)
```

Fail / Truth

Grade : F
-100
-1000



> 9

elif > 8

elif

> 7

> 6

> 5

if cgpa > 9 :

else if

elif cgpa > 8 :

elif cgpa > 7 :

else :

```
cgpa = 8.99

if cgpa > 9:
  print("S")
elif cgpa > 7:
  print("B")
elif cgpa > 8:
  print("A")
else:
  print("C")
```